Vehicle Rental System

Week 8 Practice - Question 1

Design: Customer has many Rentals. Vehicle has at most one active Rental at a time; Sedan, SUV, Truck override dailyRate().
Source note: the prompt does not define exact rates, so this example uses Sedan $45/day, SUV $70/day, Truck $100/day.
Java solution:

```
abstract class Vehicle {final String id;private boolean rented=false;Vehicle(String
i){id=i;}abstract double dailyRate();boolean available(){return !rented;}void
rent(){if(rented)throw new IllegalStateException(id+" is currently
unavailable.");rented=true;}void returned(){rented=false;}double charge(int
days){if(days<1)throw new IllegalArgumentException("Days must be positive");return
days*dailyRate();}}
class Sedan extends Vehicle {Sedan(String i){super(i);}double dailyRate(){return 45;}}
class SUV extends Vehicle {SUV(String i){super(i);}double dailyRate(){return 70;}}
class Truck extends Vehicle {Truck(String i){super(i);}double dailyRate(){return 100;}}
class Customer {final String name;Customer(String n){name=n;}}
class Rental {final Customer customer;final Vehicle vehicle;final int days;boolean
active=true;Rental(Customer c,Vehicle v,int d){customer=c;vehicle=v;days=d;}void
returnVehicle(){if(active){active=false;vehicle.returned();}}}
class RentalDesk {Rental rent(Customer c,Vehicle v,int days){if(!v.available())throw new
IllegalStateException(v.id+" is currently unavailable.");v.rent();return new
Rental(c,v,days);}}
public class Main {public static void main(String[] args){RentalDesk d=new
RentalDesk();Vehicle sedan=new Sedan("Sedan A"),suv=new SUV("SUV B");
Rental r=d.rent(new Customer("Customer 1"),sedan,3);System.out.println("Sedan A rented
successfully by Customer 1. Rental charge: $"+r.vehicle.charge(3));
try{d.rent(new Customer("Customer 2"),sedan,2);}catch(IllegalStateException
e){System.out.println(e.getMessage());}
r.returnVehicle();System.out.println("Sedan A returned by Customer 1.");Rental s=d.rent(new
Customer("Customer 3"),suv,5);System.out.println("SUV B rented successfully by Customer 3.
Rental charge: $"+s.vehicle.charge(5));}}
```

Expected result with example rates: Sedan A $135, SUV B $350. Returning the vehicle re-enables future rentals.
New categories can extend Vehicle with a different dailyRate without modifying RentalDesk.